Curator — dedup, ranking, and consolidation of swarm output

SPEC-TH-012 v0.1 DRAFT Jeremie Nunez

1 Context

The Nano-Swarm (SPEC-TH-010) produces a wide, noisy set of retrieval artifacts, some duplicated, some off-topic, some already known to the system. Before any of these are injected into the feed or promoted into a persistent source list, they must be scored for *relevance* (do they answer the research intent?) and *novelty* (do they add information we do not already have?). The Curator is the last stage of the Explorer pipeline and turns the raw artifact stream into a typed consolidation decision stream.

2 Goals

- Score each artifact on two orthogonal axes: relevance $[0, 1]$ and novelty $[0, 1]$.
- Produce a ternary decision per artifact — `inject`, `promote`, or `discard` — using deterministic thresholds.
- Assign every artifact a coarse category so downstream consumers can route it.
- Remain operational without the underlying LLM: a heuristic fallback preserves ordering semantics when the model call fails.
- Preserve input order and cardinality: exactly one `CuratedItem` out per input artifact.

3 Non-Goals

Non-Goal

This spec does not cover: the feed-injection SQL, the source-promotion logic, cross-cycle novelty memory (novelty is evaluated against what the LLM or heuristic can infer from the batch), or the prompt wording itself (implementation detail).

4 Invariants

Invariant

The Curator is total: `curate(articles)` returns a list of exactly `articles.length` items, in the same order.

Invariant

`relevanceScore` and `noveltyScore` are clamped to $[0, 1]$ for every returned item, including for heuristic-path items.

Invariant

The `action` field is one of exactly {`"inject"`, `"promote"`, `"discard"`}; `category` is one of exactly {`"MARKET"`, `"REVIEWS"`, `"DROPS"`, `"DISCOVERY"`}. Any out-of-range value from the model is coerced to `"discard"` / `"DISCOVERY"` respectively.

Invariant

An LLM failure on one batch does not contaminate other batches: the Curator falls back to the heuristic scorer for that batch only and continues.

Invariant

Given *relevance* ≥ 0.7 and *novelty* ≥ 0.5, the heuristic path yields `action = "inject"` (or stronger: `"promote"` if relevance ≥ 0.8 and entity count ≥ 5).

5 Interface

The Curator consumes the artifact type produced by SPEC-TH-010 and emits a decision envelope.

```
export interface CuratedItem {
  url: string;
  title: string;
  body: string;
  relevanceScore: number;    // [0,1]
  noveltyScore: number;      // [0,1]
  action: "inject" | "promote" | "discard";
  category: "MARKET" | "REVIEWS" | "DROPS" | "DISCOVERY";
  reason: string;            // <= 200 chars, 1 sentence
  entities: ExtractedEntities;
}

export class ExplorerCurator {
  curate(articles: RetrievalArtifact[]): Promise<CuratedItem[]>;
}
```

6 Scenarios

Scenario: Nominal batch scoring

Given 25 retrieval artifacts and a healthy LLM transport
When `curate` is invoked
Then the artifacts are split into batches of at most 10
And each batch is scored with one LLM call
And the returned list has exactly 25 items, in the original input order
And every score is clamped to $[0, 1]$

Scenario: LLM unavailable for one batch

Given three batches where the second one causes the LLM call to reject
When `curate` is invoked
Then batches 1 and 3 are scored by the LLM
And batch 2 is scored by the heuristic fallback
And the final list preserves the original input order and length

Scenario: Malformed model output

Given the LLM returns JSON where `action` is `"maybe"` and `category` is `"OTHER"`
When the batch is parsed
Then those fields are coerced to `"discard"` and `"DISCOVERY"`

And the item is still returned (not dropped)

Scenario: Heuristic promotion

Given an artifact with rich wine entities (≥ 5), concrete data points, and body ≥ 1500 chars
When the heuristic path runs on it
Then `relevanceScore` is at least 0.8
And `action` is `"promote"`

Scenario: Empty input

Given an empty artifact list
When `curate` is invoked
Then the result is `[]`
And no LLM call is made

7 Acceptance Criteria

Acceptance Criterion AC-1

For every input artifact there is exactly one output item with the same `url`, in the same position.

Acceptance Criterion AC-2

`relevanceScore` and `noveltyScore` are in $[0, 1]$ for every returned item.

Acceptance Criterion AC-3

`action` and `category` always belong to their declared enumerations; unknown values from the model are coerced to `"discard"` / `"DISCOVERY"`.

Acceptance Criterion AC-4

A forced LLM failure on one batch results in that batch being heuristic-scored; the other batches are unaffected.

Acceptance Criterion AC-5

An input of size 0 returns `[]` and triggers zero LLM calls.

Acceptance Criterion AC-6

For an artifact with ≥ 5 entities, ≥ 1 data point, and body length ≥ 1500, the heuristic path yields `action` $\in$ `{"inject","promote"}`.

Acceptance Criterion AC-7

`reason` is a non-empty string no longer than 200 characters for every item.

Acceptance Criterion AC-8

Batches submitted to the LLM contain at most 10 artifacts; a mock LLM transport asserts this bound.

8 Traceability

AC	Test file	Test name
AC-1	`packages/thalamus/tests/curator.order.spec.ts`	`preserves input order and cardinali`
AC-2	`packages/thalamus/tests/curator.clamp.spec.ts`	`clamps scores to [0,1]`
AC-3	`packages/thalamus/tests/curator.enum.spec.ts`	`coerces unknown enum values`
AC-4	`packages/thalamus/tests/curator.fallback.spec.ts`	`falls back to heuristic on LLM fail`
AC-5	`packages/thalamus/tests/curator.empty.spec.ts`	`empty input returns empty output`
AC-6	`packages/thalamus/tests/curator.heuristic.spec.ts`	`promotes entity-rich artifacts`
AC-7	`packages/thalamus/tests/curator.reason.spec.ts`	`reason is always bounded`
AC-8	`packages/thalamus/tests/curator.batching.spec.ts`	`batches at most 10 artifacts per LL`

9 Open Questions

- Should novelty be evaluated against a cross-cycle memory (e.g. prior URLs, prior entity fingerprints) rather than from the LLM alone?
- Should the `category` enumeration be pluggable per-domain, or remain fixed at the Explorer level?
- Should the heuristic thresholds (0.7 / 0.5 / 0.8) be configurable, or fixed as part of the contract?